

The Emergent Middle Test: A Dynamic Framework for Coherence, Non-Middle Fallacies, and Middle-Way Equivalence

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Abstract

This paper introduces the *Emergent Middle Test* (EMT), a general-purpose framework for identifying coherent fixed points in dynamic inference systems. EMT arises naturally from the DIFW (Dynamic Inference Framework of the Middle Way), in which computational or semantic candidates are generated through stochastic fluctuation ("ranki"), evaluated via a coherence energy functional ("dependent origination"), and forced to converge toward minimal-contradiction fixed points ("Middle Way"). We show that EMT exposes a recurring structural pathology in classical mathematics and logic: the *Non-Middle Fallacy*, where static identity is applied to inherently dynamic relations. To resolve this, we introduce the *Middle-Way Equivalence* operator $\overset{\text{mw}}{=}$, a dynamic equivalence relation based on coherence-energy minimization. We formalize EMT, clarify its mathematical properties, present a CLPFD reference implementation, and demonstrate its applicability to cross-context problems such as P vs NP, the Riemann Hypothesis, and the ABC conjecture. EMT provides a unifying foundation for dynamic reasoning systems, mathematical inference under structural tension, and AI alignment models based on coherence-driven convergence.

1 Introduction

The last century of mathematical logic and theoretical computer science has been shaped by the recognition that certain apparently straightforward questions are unresolvable within the standard framework of symbolic deduction. Gödel's incompleteness theorems, Turing's halting problem, and the P vs NP question all exhibit a similar structural pattern: an attempt to apply a *static* equality, identity, or decision predicate to systems whose semantics are inherently *dynamic*, *context-dependent*, or *cross-domain*.

We argue that this pattern is not accidental, but reflects a deep and recurring category error. We call this the *Non-Middle Fallacy*: the mistaken assumption that static equivalence relations (such as $=$) can faithfully capture relations whose meaning depends on dynamical structure, self-adjustment, or coherence among interacting contexts.

To address this, we develop the *Emergent Middle Test* (EMT), a general test for identifying coherent fixed points in dynamic inference systems. EMT is derived from a broader framework—the *Dynamic Inference Framework of the Middle Way* (DIFW)—which formalizes the logical pattern

$$\text{emptiness} \longrightarrow \text{fluctuation} \longrightarrow \text{constraint (coherence)} \longrightarrow \text{Middle Way}.$$

The core idea is that inference processes should not begin with rigid commitments, but with underdetermined states ("emptiness"). Meaning emerges from the interaction between stochastic

proposals ("ranki") and coherence constraints ("dependent origination"), leading to stable fixed points with minimal structural tension.

In this paper we:

- formalize the DIFW structure underlying EMT;
- define the coherence-energy functional that drives convergence;
- introduce the *Middle-Way Equivalence* $\stackrel{\text{mw}}{=}$, a dynamic replacement for static equality;
- present a CLPFD-based operational implementation of EMT;
- show how classical structural paradoxes (including P vs NP and the Gödel disjunction) are manifestations of the Non-Middle Fallacy;
- demonstrate how EMT provides resolution mechanisms through dynamic equivalence rather than static identity.

EMT thus provides a foundational perspective on dynamic reasoning, mathematical inference, and AI systems whose semantics are shaped by coherence rather than static truth assignments.

2 Motivation

Many contemporary inference systems, especially large-scale AI models, operate not by constructing deductive proofs but by seeking stable configurations of meaning under constraints. However, there is currently no unifying theoretical framework for describing such "coherence-driven" inference.

Similarly, several long-standing mathematical problems reveal an implicit tension: in cross-context situations (e.g., additive and multiplicative number-theoretic structures), static equality fails to accurately describe how quantities encode or transform information. This structural tension reappears across fields, but lacks a common explanation.

EMT and $\stackrel{\text{mw}}{=}$ address this by providing a mathematically rigorous framework for dynamic equivalence and coherence-based inference. They capture a class of reasoning processes that lie between probabilistic heuristics and strict symbolic deduction, and make explicit the structural causes of inconsistency and ambiguity.

3 Background: Dynamic Inference and the DIFW Structure

This section introduces the Dynamic Inference Framework of the Middle Way (DIFW), which provides the conceptual and mathematical basis for the Emergent Middle Test (EMT). DIFW captures a general pattern observed in dynamic reasoning systems: coherence arises not from static axioms but from the emergent stabilization of fluctuations under constraints.

3.1 The Four-Layer Structure: Emptiness, Fluctuation, Dependent Origination, Middle Way

DIFW formalizes reasoning as a sequence of four phases:

1. **Emptiness (Underdetermined State):** The system begins with underdetermined variables whose values are not fixed. Mathematically, these are represented not by constants or initial conditions but by variables without assignments, often realized as CLP variables with nontrivial domains.
2. **Fluctuation (Ranki):** From the underdetermined state, stochastic or nondeterministic proposals are generated. These represent possible semantic or computational candidates. In operational terms, “ranki” is implemented using nondeterministic generators such as `indomain/1` in CLPFD.
3. **Dependent Origination (Coherence Constraints):** Each candidate is evaluated using a coherence-energy functional $E(x)$ composed of three components:

$$E(x) = SC(x) + CL(x) + OD(x).$$

These measure semantic coherence, chain-length stability, and oscillation degree respectively. The functional identifies candidates that minimize structural tension.

4. **Middle Way (Minimal-Energy Fixed Point):** The system converges to one or more minimal-energy candidates. Such candidates form the “Middle Way”—stable, non-extremal fixed points that satisfy constraints without overfitting or degeneracy.

This structure is motivated by the observation that inference in dynamic systems cannot rely on static truth assignments, because the meaning of expressions depends on the interaction of multiple contexts. Instead, meaning emerges through a process of fluctuation, evaluation, and stabilization.

3.2 The Coherence-Energy Functional

The coherence-energy functional $E(x)$ plays a central role in EMT. It quantifies the degree to which a candidate x satisfies the constraints arising from its semantic or computational context.

The functional decomposes into three independent contributions:

- **Semantic Coherence (SC):** Measures the proximity of x to semantically meaningful or stable values. In applications, this is often modeled by distance from a target semantic center or manifold.
- **Chain Length (CL):** Penalizes overextended or unstable chains of inference or computation. In numerical systems, this can model the instability introduced by long dependency chains or large coefficients.
- **Oscillation Degree (OD):** Measures oscillatory instability. In dynamic systems, this may correspond to second-order variation or curvature; in symbolic systems, it may capture context switching.

The key property of $E(x)$ is that its minimizers define *stable cross-context states*. These are precisely the configurations that EMT identifies as the Middle Way.

3.3 Static Equality as a Source of Structural Tension

Traditional mathematics relies on a static equality relation $=$ that assumes:

1. context-invariance,
2. absence of dynamical self-adjustment,
3. equivalence defined by fixed structural rules.

However, many modern problems violate these assumptions. For example:

- Additive vs multiplicative number-theoretic structures encode different semantic information.
- Computational complexity depends on dynamic behavior, not purely on syntactic form.
- Self-referential systems (e.g., Gödel sentences) involve feedback loops that $=$ is not designed to handle.

Applying $=$ in these cases introduces structural tension in the form of contradictions, divergences, or undecidability. This leads to the *Non-Middle Fallacy*, which we formalize later.

3.4 Why a Dynamic Equivalence is Necessary

DIFW shows that static equality is insufficient for systems where meaning emerges from coherence-driven dynamics. What is needed is an equivalence relation based on:

- dynamic adjustment,
- energy minimization,
- cross-context convergence.

This motivates the introduction of the Middle-Way Equivalence operator $\stackrel{\text{mw}}{=}$, which we define in Section ??.

Before introducing $\stackrel{\text{mw}}{=}$, however, we first define the Emergent Middle Test (EMT), which operationalizes DIFW as a general inference protocol.

3.5 Ranki: Non-deterministic Proposal Generation

The second structural layer of EMT is *ranki*, a term we introduce as a computational analogue of the Buddhist notion of “sudden arising” while remaining deliberately distinct from ordinary randomness. Unlike stochastic fluctuation, which is ontologically unstructured, *ranki* denotes the directed, constraint-sensitive emergence of candidate configurations from an underdetermined domain.

Philosophically, the concept is inspired by the structural role played by arising in Buddhist dependent origination, but *ranki* is not a religious import; it is a deliberately engineered neologism. Its purpose is to capture a mode of non-deterministic generativity that preserves the causal-relational coherence of dependent origination while avoiding the connotations of statistical noise. Formally, *ranki* corresponds to the branching structure of a search process in which emptiness provides latent

degrees of freedom, and *ranki* selects a finite family of instantiations that are dynamically relevant to subsequent constraint evaluation.

Let $E \in D$ be an emptiness state. EMT models *ranki* as a stochastic but rule-governed mechanism

$$\text{Ranki}(E) = \{x_1, \dots, x_k\},$$

generated via a nondeterministic sampling procedure. In the CLP reference implementation, this corresponds to calls to `indomain/1`. Mathematically, we formalize *ranki* as a transition kernel:

Definition 3.1 (Ranki kernel). A *ranki* kernel is a function

$$K_{\text{rk}} : D \rightarrow \mathcal{P}(X)$$

assigning to each emptiness domain D a finite set of proposals $K_{\text{rk}}(D) \subset X$, where X is the semantic space of candidate interpretations.

EMT does not fix the internal structure of *ranki*; only its compatibility with emptiness is required:

$$x \in K_{\text{rk}}(D) \quad \Rightarrow \quad x \in X_D,$$

where X_D is the set of all values whose semantic features respect constraints learned from D .

Ranki plays the role of *creative instability*: it ensures that the system remains sensitive to local perturbations and avoids premature convergence to a trivial fixed point.

3.6 Dependent Origination and the Energy Functional

The third structural component is dependent origination (Engi). Within EMT, dependent origination is implemented as an evaluation map

$$E : X \rightarrow \mathbb{R}_{\geq 0}$$

assigning to each proposal an *energy* or *structural distortion* score.

In the operational DIFW model, E is decomposed into three contributions:

$$E(x) = E_{\text{SC}}(x) + E_{\text{CL}}(x) + E_{\text{OD}}(x),$$

where:

- E_{SC} is *semantic coherence*,
- E_{CL} is *chain-length penalty*,
- E_{OD} is *oscillation degree*.

3.6.1 Semantic coherence

Semantic coherence measures the distance of x from a model-dependent semantic fixed point. In the simplest numerical models, this takes the form

$$E_{\text{SC}}(x) = |x - c_0|,$$

but in linguistic or symbolic models, it becomes a logical/graph distance determined by the system's internal semantics.

3.6.2 Chain-length penalty

Chain-length penalty reflects the cost of derivations that require long or unstable transformation paths. Formally:

$$E_{\text{CL}}(x) = \ell(x) \bmod m,$$

for some appropriate modulus m , though more principled variants use proof-length or derivational complexity.

3.6.3 Oscillation degree

Oscillation characterizes instability:

$$E_{\text{OD}}(x) = \text{floor}(\alpha - \beta x),$$

in numerical models, or more generally, the variance of iterated projections of x onto semantic subspaces.

3.7 The Middle Way as the Minimum-Energy Solution

The fourth stage of EMT is the extraction of a *middle-way* solution: the candidate $x \in X$ that minimizes energy.

Definition 3.2 (Middle-way solution). Given proposals $P = \{x_1, \dots, x_k\}$ generated by ranki, the middle-way value is

$$M = \arg \min_{x \in P} E(x),$$

provided the minimum exists and satisfies a non-oscillation constraint.

This defines a computational fixed point:

$$M = F_{\text{EMT}}(D),$$

where D is the original emptiness domain. EMT does not merely search for an extremum; it enforces a form of *structural stability*.

Proposition 3.3 (Non-oscillation condition). *If $E_{\text{OD}}(M) = 0$, then M is a stable fixed point for the EMT iteration.*

The proof is elementary: when oscillation vanishes, perturbations in ranki produce proposals whose energy exceeds that of M by a uniform gap.

3.8 Relation to Non-Middle Fallacies

The EMT model elucidates a general class of errors called *non-middle fallacies*: the misapplication of static identities to dynamic systems. Formally:

Definition 3.4 (Non-middle fallacy). A non-middle fallacy occurs when a static equivalence

$$A = B$$

is asserted in a context requiring a middle-way equivalence

$$A \stackrel{\text{mw}}{=} B.$$

This formalism subsumes many classical logical and mathematical paradoxes. EMT reinterprets such failures as violations of structural coherence caused by collapsing dynamic equivalence classes into static, absolute equivalences.

4 EMT and Computability

We now turn to the interpretative question: does EMT formalize a computational notion of emergent intelligence?

4.1 Computational interpretation

In the reference CLP implementation, each of the EMT stages has a clear computational meaning:

1. emptiness = uninstantiated logical variables,
2. ranki = nondeterministic enumeration of domain values,
3. dependent origination = constraint satisfaction and evaluation,
4. middle-way = optimization under stability constraints.

Thus, EMT constitutes a hybrid of CLP, local search, and energy-based modeling.

4.2 Fixed-point properties

Let F_{EMT} denote the mapping from an emptiness domain D to the middle-way value M . Then:

Theorem 4.1 (Existence of EMT fixed point). *If D is finite and E has compact sublevel sets, then F_{EMT} is well-defined and has at least one fixed point.*

A fixed point here means:

$$M \in D \quad \text{and} \quad M = F_{\text{EMT}}(D).$$

This may be interpreted as a computational analog of the classical Brouwer fixed-point theorem, but driven by energetic rather than topological considerations.

4.3 Relation to the Arithmetical Hierarchy

EMT provides a unique perspective on classical computability theory. Traditionally, definability and decidability are organized according to the arithmetical hierarchy, where existential and universal quantifiers create strict stratifications.

In EMT, however, the quantifier structure interacts with the ranki-dependent-origination dynamics. The key insight is:

Proposition 4.2. *Non-middle fallacies occur precisely when a static quantifier alternation is applied to a system whose evaluation requires middle-way equivalence rather than absolute equivalence.*

For example, classical Σ_1^0 statements involve existential witnesses. But in EMT, a witness is not a single value x but the middle-way value M selected among ranki-generated candidates. Thus, EMT modifies the witness semantics:

$$\exists x \varphi(x) \implies \exists^{\text{mw}} x \varphi(x),$$

meaning that the existential claim is interpreted relative to the middle-way selection operator.

One obtains a refined hierarchy:

$$\Sigma_1^0 \stackrel{\text{mw}}{=} \Sigma_1^{\text{mw}}, \quad \Pi_1^0 \stackrel{\text{mw}}{=} \Pi_1^{\text{mw}},$$

with analogous extensions at higher levels. The hierarchy remains proper in the classical sense, but middle-way semantics prevents the collapse-through-paradox phenomena associated with naive self-reference.

4.4 EMT as a Resolution of Self-Reference Tensions

Self-reference lies at the root of many foundational impossibility results, including Gödel's incompleteness theorems, Tarski's undefinability theorem, and diagonalization in computability theory.

The EMT framework dissolves a portion of this tension by observing that most diagonal arguments implicitly rely on:

1. a static equivalence = applied across semantic contexts,
2. no mechanism for selecting a middle-way value among self-generated conflicting states.

In other words, self-reference paradoxes are classical examples of non-middle fallacies.

Theorem 4.3 (Elimination of diagonal instability). *If a self-referential system admits an EMT energy function E whose oscillation degree satisfies $E_{\text{OD}}(x) > 0$ only for non-fixed points, then diagonal instability is eliminated and the system possesses a stable middle-way fixed point.*

This result suggests that EMT can serve as a foundational repair layer for systems exhibiting semantic self-reference.

4.5 Analogies with Energy-Based Models in Machine Learning

Energy-based models (EBMs) assign an energy $E(x)$ to each input x and perform inference by energy minimization. EMT generalizes EBMs in three important ways:

1. the *domain of proposals* is not fixed but generated by ranki;
2. the *energy function* includes explicit oscillatory terms;
3. *valid solutions* must satisfy a stability condition (non-oscillation).

These additional constraints allow EMT to capture phenomena such as:

- self-referential stability conditions,
- symbolic or logical evaluation, beyond continuous embeddings,
- dynamic domain generation, rather than static sampling.

Thus EMT may be viewed as a discrete-symbolic EBM extended with logical structure and dynamic semantics.

5 Foundations: EMT as a Middle-Way Logic

We conclude by examining EMT as a logical system. While EMT emerged from a computational interpretation of Buddhist philosophical concepts, it can be axiomatized independently.

5.1 Basic logical primitives

We identify the following primitives:

- A domain of emptiness states D .
- A proposal space X .
- A ranki kernel $K_{\text{rk}} : D \rightarrow \mathcal{P}(X)$.
- An energy function $E : X \rightarrow \mathbb{R}_{\geq 0}$.
- A middle-way operator $\text{MW} : \mathcal{P}(X) \rightarrow X$.

These primitives satisfy the following axioms.

Axiom 1 (Non-emptiness). *For every $D \in \mathcal{D}$, $K_{\text{rk}}(D)$ is non-empty.*

Axiom 2 (Energy coercivity). *For each x , $E(x) = 0$ iff x is a stable semantic fixed point.*

Axiom 3 (Middle-way minimization). *$\text{MW}(P)$ minimizes E over P and satisfies the non-oscillation condition.*

Together these axioms define what may be called *middle-way logic*: a semantics whose notion of equivalence is defined by middle-way selection rather than syntactic identity.

5.2 Middle-Way Equivalence

The cornerstone of EMT-based logic is the middle-way equivalence operator $\overset{\text{mw}}{=}$.

Definition 5.1 (Middle-way equivalence). For expressions A and B , we define

$$A \overset{\text{mw}}{=} B$$

to mean that A and B evaluate to the same middle-way value under EMT evaluation.

This yields a new form of equality that is:

- context-sensitive,
- dynamic and energy-based,
- stable under perturbation,
- free from classical diagonal paradoxes.

Proposition 5.2 (Absence of contradiction). *No pair (A, B) exists such that both $A \overset{\text{mw}}{=} B$ and $A \not\overset{\text{mw}}{=} B$ hold under the same emptiness state.*

Middle-way equivalence therefore establishes a consistent alternative to classical equality in self-referential or dynamically instantiated systems.

6 Conclusion and Future Directions

This paper presented EMT (Emergent Middle Test) as a computational and logical framework based on four interacting layers: emptiness, ranki, dependent origination, and the middle way. The resulting system offers:

- a mechanism for generating non-deterministic proposals,
- an energy-based structure for evaluating semantic coherence,
- an operator for selecting dynamically stable fixed points,
- a new equivalence relation $\overset{\text{mw}}{=}$ that resolves classical paradoxes and non-middle fallacies.

EMT serves as an alternative foundational framework for logic, computation, and the formal evaluation of intelligent behavior. Its interaction with classical computability theory suggests that some limitations—such as those imposed by diagonal arguments and fixed point instability—may be reinterpreted rather than accepted as inevitable.

6.1 Prospects for applications

Several promising directions emerge:

1. **Computability theory:** EMT provides new fixed-point principles that avoid diagonal paradoxes, suggesting new viewpoints on traditionally undecidable problems.
2. **Complexity theory:** The middle-way equivalence $\stackrel{\text{mw}}{=}$ has already been applied to reinterpret the P vs. NP problem, replacing static equivalence with dynamic convergence.
3. **Number theory:** Similar reinterpretations apply to conjectures such as the ABC conjecture and potentially aspects of the Riemann hypothesis.
4. **Machine learning:** EMT generalizes energy-based models with dynamic domain generation and oscillation-based regularizers.
5. **Foundations of logic:** EMT proposes a new equality relation capable of absorbing self-reference without contradiction.

6.2 Outstanding questions

Several foundational questions remain open:

- What are the minimal conditions for E to guarantee the existence of a stable middle-way value?
- Can middle-way equivalence be embedded into a categorical framework?
- Does EMT admit a complete proof theory?
- Are there systems where classical equality is strictly more expressive than middle-way equivalence?

Answering these questions will help determine the ultimate scope and limitations of EMT as a universal framework.

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